Compare cones, round counts

Pinecones

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. Little counts 47 open cones and 74 closed cones on two trays. Fill the circle with >, <, or =. Which count is greater?

47 ○ 74

Show your thinking.

2. Big records 32 visible scales. What is this count rounded to the nearest ten?

32 → nearest ten: ___

Show your thinking.

3. One notebook shows 68 dry cones; another shows 59 damp cones. Fill the circle with >, <, or =. Which count is greater?

68 ○ 59

Show your thinking.

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Pinecones

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4. Little draws 85 moisture marks beside the cone pictures. What is the count rounded to the nearest ten?

85 → nearest ten: ___

Show your thinking.

5. Big and Little each count 26 scales on one side of a cone. Fill the circle with >, <, or =. What is the shared count?

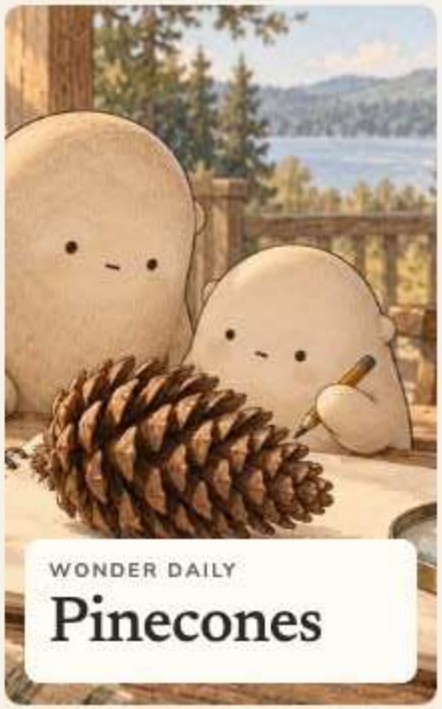
26 ○ 26

Show your thinking.

6. A page shows 56 open scales. What is the count rounded to the nearest ten?

56 → nearest ten: ___

Show your thinking.



WONDER DAILY

Pinecones

Make a little book



1 Fold it into 8 boxes. Open it flat.



2 Fold short end to short end. Cut the thick line. Open it.



3 Fold the long way, pictures out. Push the ends in.



4 Fold into a book. Page 1 goes on the front!



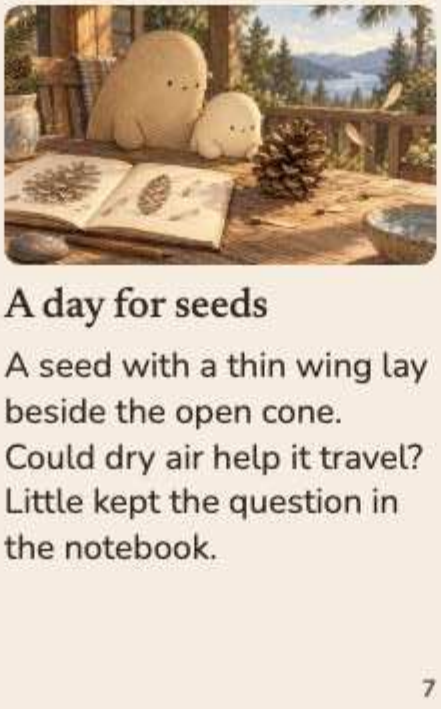
Room for seeds

The dry cone had gaps between its scales. Seeds could slip through those spaces. Little drew its wide, open shape.



A woody spiral

Little found a woody seed case. Its scales overlapped in a spiral. Big called it a pinecone. What was hiding inside?



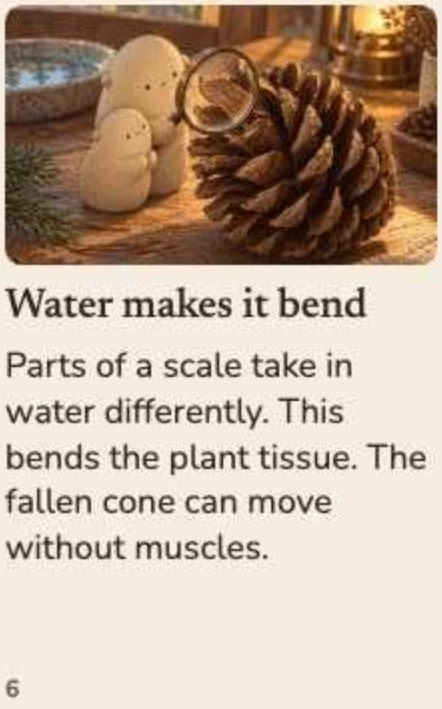
A day for seeds

A seed with a thin wing lay beside the open cone. Could dry air help it travel? Little kept the question in the notebook.



A slow closing

The wet cone took in a little water. Its scales bent inward and closed the gaps. Little watched its shape grow narrow.



Water makes it bend

Parts of a scale take in water differently. This bends the plant tissue. The fallen cone can move without muscles.



Two dishes, one clue

Little drew the open cone and the closed cone. When the damp cone dried, spaces began to appear between its scales again.

A small change, in words

Pinecones · Curiosity day

Name _____ Date _____

Give a clear command for watching a cone.



Room between scales



A cone closes



Two dishes, one clue

Draw an open cone beside a closed cone. Show the gaps between the scales.

pinecone · scale · dry · wet · open · close · seed · slowly

Watch … tell which cone to observe

Compare … tell which shapes to compare

Draw … tell which details to include

Does your command begin with an action word, such as Watch, Compare, or Draw?

Pinecones

01 What is a pinecone made of?



Scales stack in a spiral around a pinecone.

Look closer. Follow a spiral around the cone.

A little explanation. A pinecone is a woody seed case with many overlapping parts called scales. Seeds grow tucked beside the scales. The scales wind around the cone in spirals.

Try this. Count the scales you can see on one side, then draw one spiral.

02 Why are there gaps in a dry cone?



Dry scales open, leaving room for seeds to leave.

Look closer. Look between the spread-out scales.

A little explanation. When many common pinecones dry, their scales bend outward. The open spaces let ripe seeds leave. A light wing helps many pine seeds catch the breeze.

Try this. Set a dry cone on a dish and sketch its outline. Leave room beside your drawing for a second shape.

03 What changes when a cone gets wet?



Wet scales bend inward and close the gaps.

Look closer. Compare the close scales with your open-cone drawing.

A little explanation. Wet or humid conditions add water to the scales. Parts of each scale take in water differently, so the woody tissue bends inward. The cone closes. A fallen cone can still do this without muscles.

Try this. Use two fallen cones or cones from a craft bin. Put one in a dry dish and briefly wet the other in a shallow dish of water. With Big beside you, check their scales over time and draw both shapes. Collect fallen brown cones with Big nearby; leave green cones on branches, keep feet on the ground, and wash hands afterward.

04 Why wait for dry air?



Dry weather can help winged seeds drift away.

Look closer. Find the thin wing beside a seed.

A little explanation. Winged seeds can drift on a dry breeze. When wet, they can cling to a scale instead. Closing in damp weather helps keep seeds inside until conditions are better for leaving. Different kinds of pinecones do not all open in exactly the same way.

Try this. Let the damp cone dry on an empty dish. Check it later. What do you think the scales will do?